PROJECT MODULE
TASK



WIRESHARK PACKET CAPTURE

CYBERSECURITY & ETHICAL HACKING PROJECT TASKS

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WIRESHARK PACKET CAPTURE

Background

Wireshark is a free tool that captures the packets moving across a network and lets you read them one by one. Network engineers use it to find problems and security teams use it to spot suspicious traffic.

When you open a website, your laptop first asks a DNS server for the IP address of that website. It then connects to that IP address and starts a secure TLS connection. In this lab you will capture your own Wi-Fi traffic while you open a website, and read this information from the packets.

Task

Capture your own Wi-Fi traffic with Wireshark on your Windows laptop while you open networkwalks.com, then find the six answers below in the packets.

Website: https://networkwalks.com

Questions to answer:

1. What is the IP address of your own laptop?
2. What is the IP address of your DNS server?
3. What is the IP address of the website networkwalks.com?
4. Which website name (Server Name) is sent in the TLS Client Hello?
5. Which server port is used for the secure connection?
6. Which TLS version does the website use?

Submit one screenshot of Wireshark showing the DNS answer with the website IP, and your answers to the six questions.

Solution

STEP 1

Close all browser windows. Open Command Prompt and run this command to clear the DNS cache:

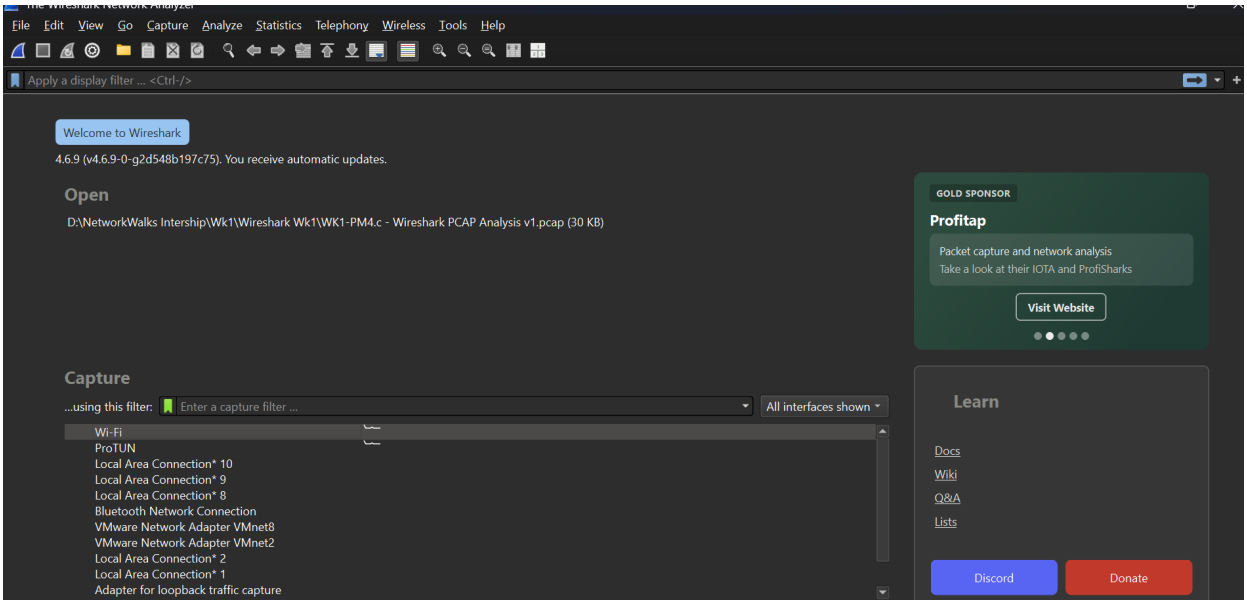
```
ipconfig /flushdns
```

You should see Successfully flushed the DNS Resolver Cache. Without this step your laptop may remember the website and skip the DNS query.



STEP 2

Open Wireshark. On the start screen, double-click the adapter that shows a moving line (usually Wi-Fi). The capture starts at once.

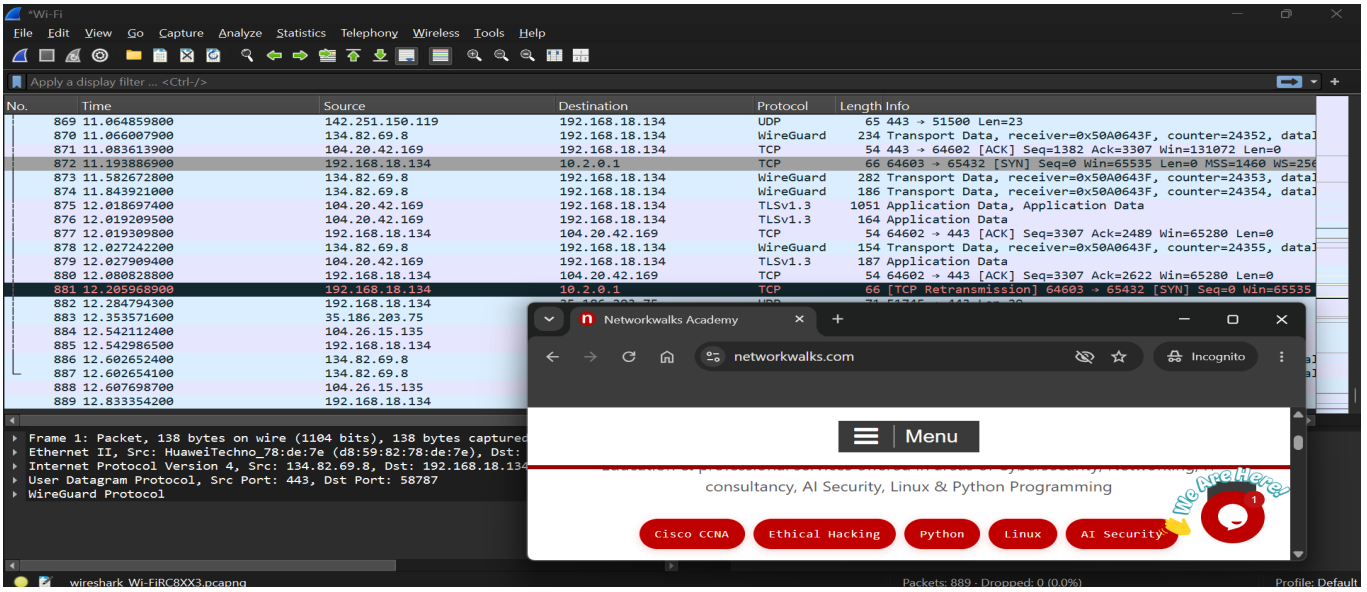


STEP 3

Open a new Incognito window in Chrome (Ctrl + Shift + N) and go to the website. Wait until the page loads fully.

https://networkwalks.com

Come back to Wireshark and stop the capture with the red square button.

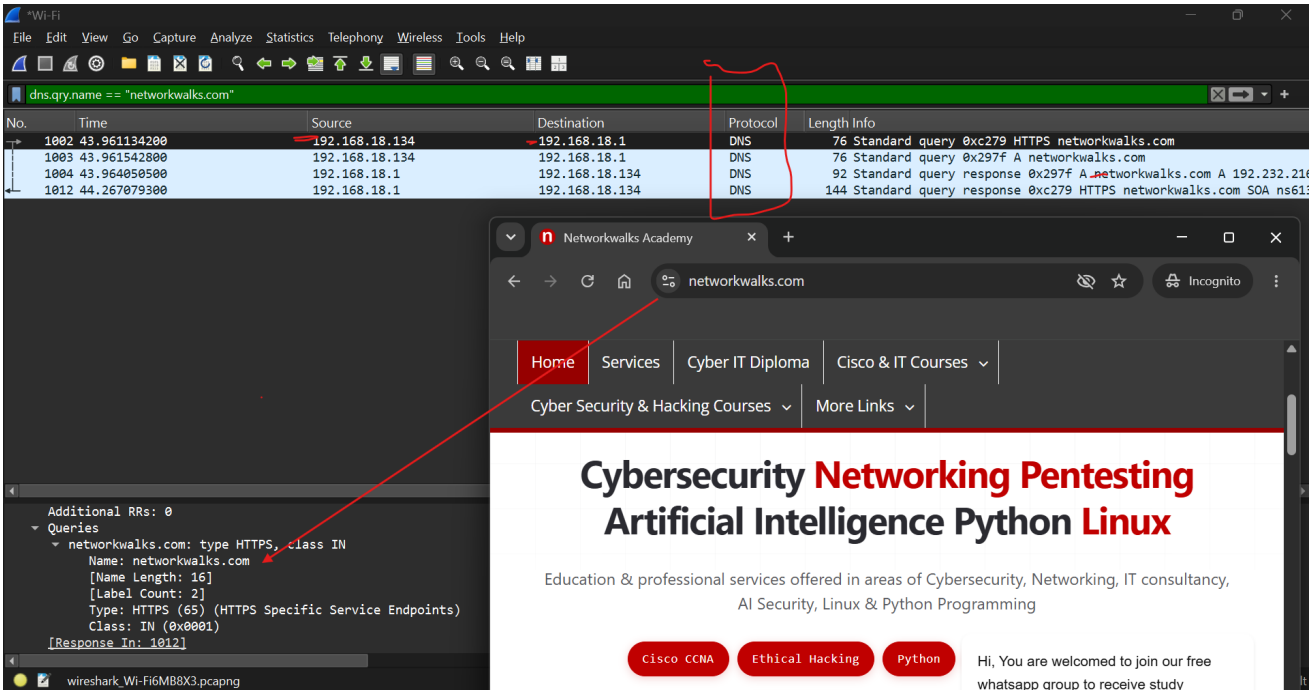


STEP 4

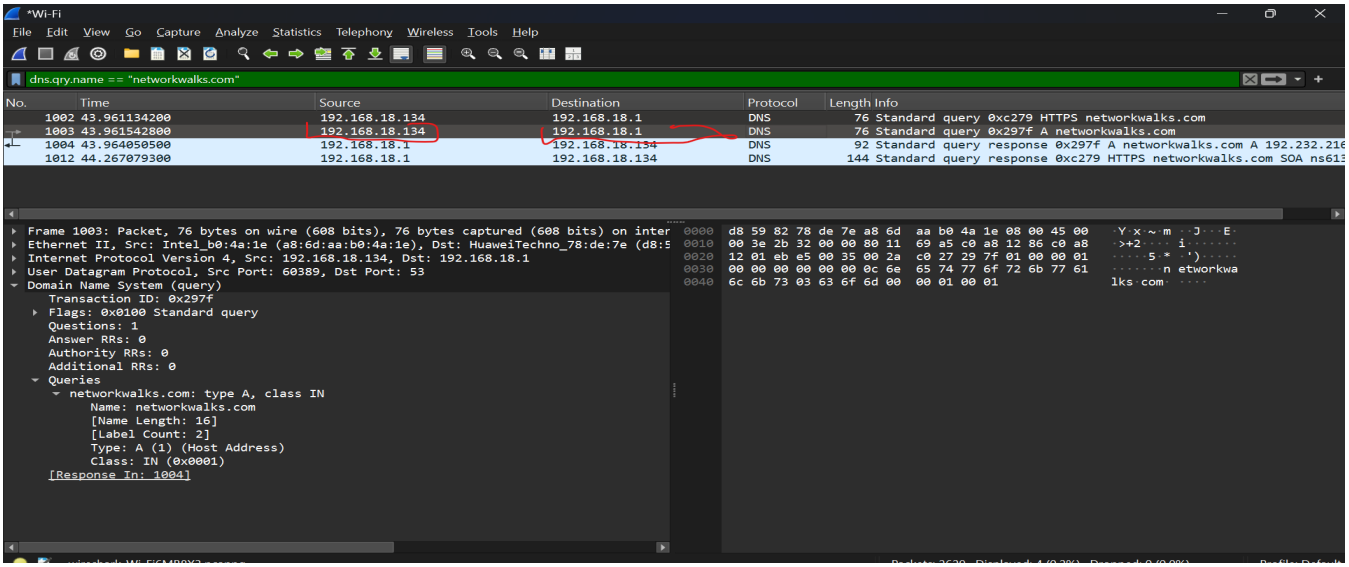
Type this in the display filter bar and press Enter:

```
dns.qry.name == "networkwalks.com"
```

You should see two packets, a query and a response.



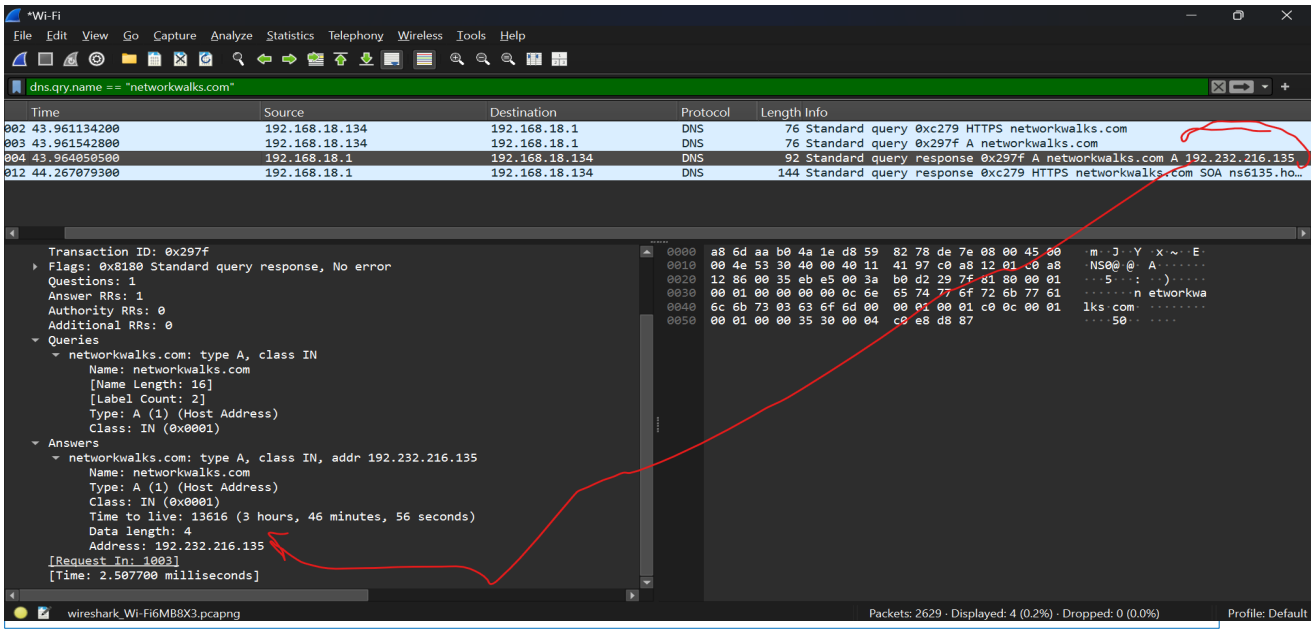
Click the query. The Source is your laptop and the Destination is your DNS server.



Answer1: Laptop IP is 192.168.18.134

Answer2: DNS Server IP is 192.168.18.1

Now click the response. In the middle pane expand Domain Name System (response), then Answers. The Address is the website IP.



Answer3: Website IP is 192.232.216.135

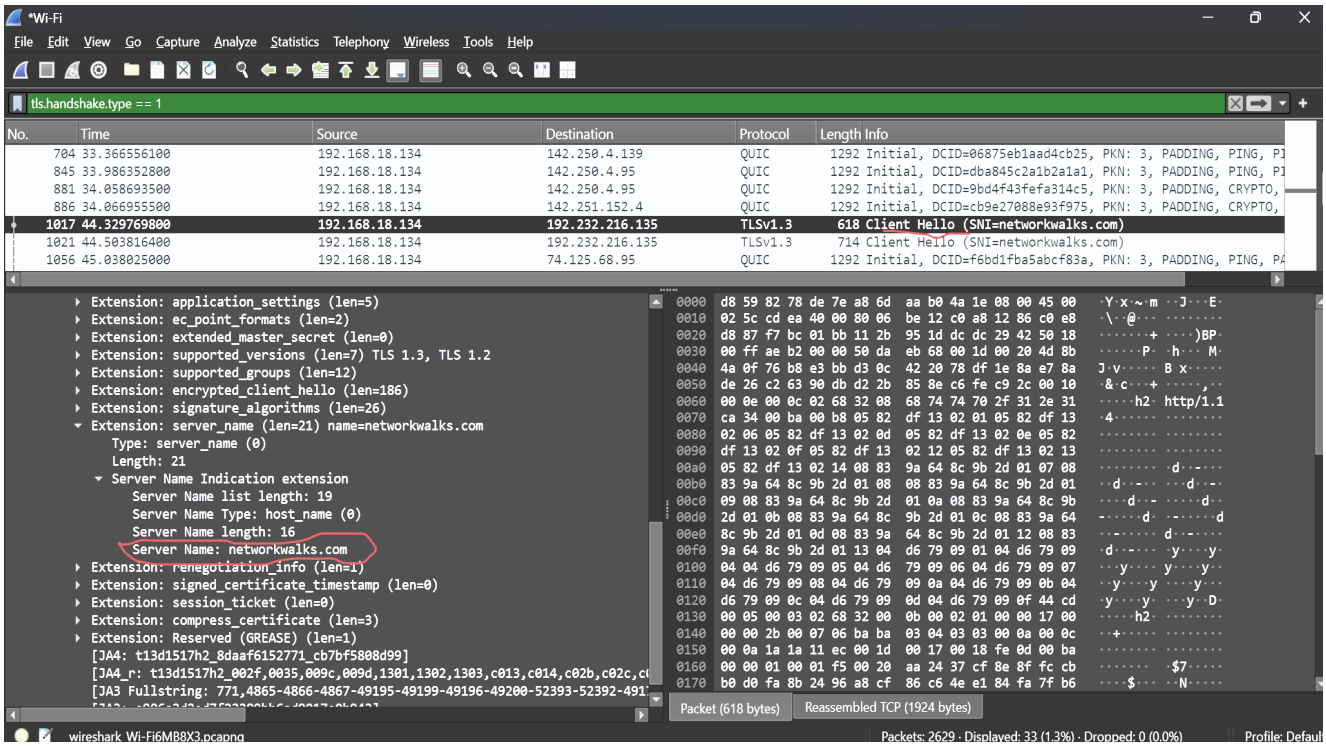
Tip: If no packet shows, close the browser fully, run ipconfig /flushdns again and capture again. Also check that Chrome Secure DNS is off (Settings > Privacy and security > Security), because it hides the DNS query.

STEP 5

Use this filter:

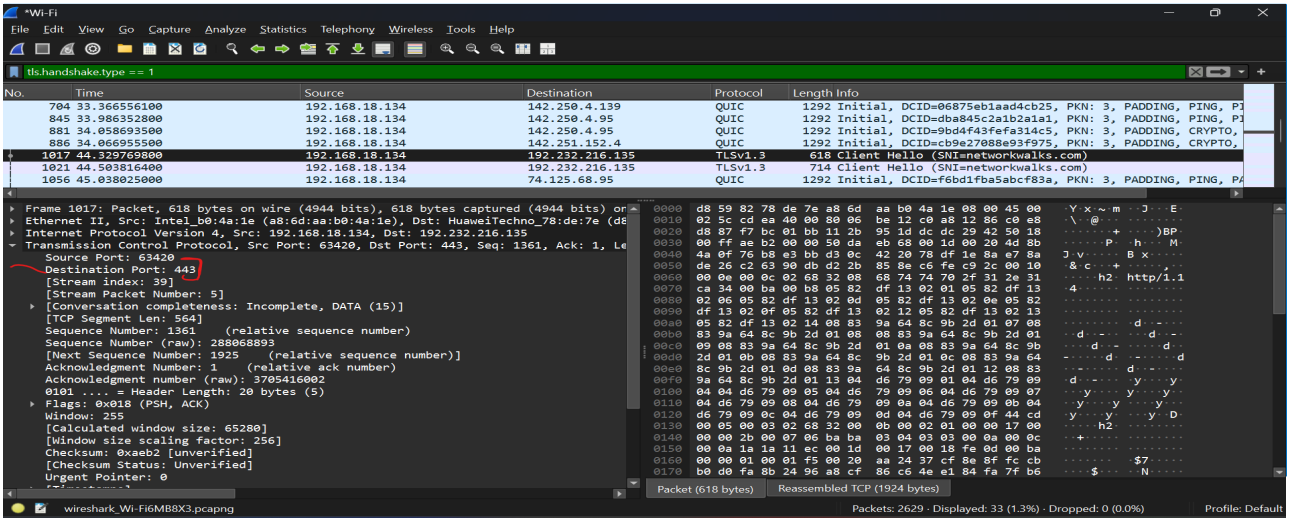
`tls.handshake.type == 1`

This shows the Client Hello, the first packet of the secure connection. Click the one that says SNI=networkwalks.com. Its Destination is the website IP. Expand Transport Layer Security > Handshake Protocol: Client Hello > Extension: server_name.



Answer4: Server Name: networkwalks.com

Then expand the Transmission Control Protocol line. The Destination Port is the Server port.



Answer5: Server Port is 443

Tip: If the Protocol column says QUIC, this is normal. Chrome uses QUIC (HTTP/3) on port 443 for some sites. The filter still works. If you see no packet, try the same steps in another browser.

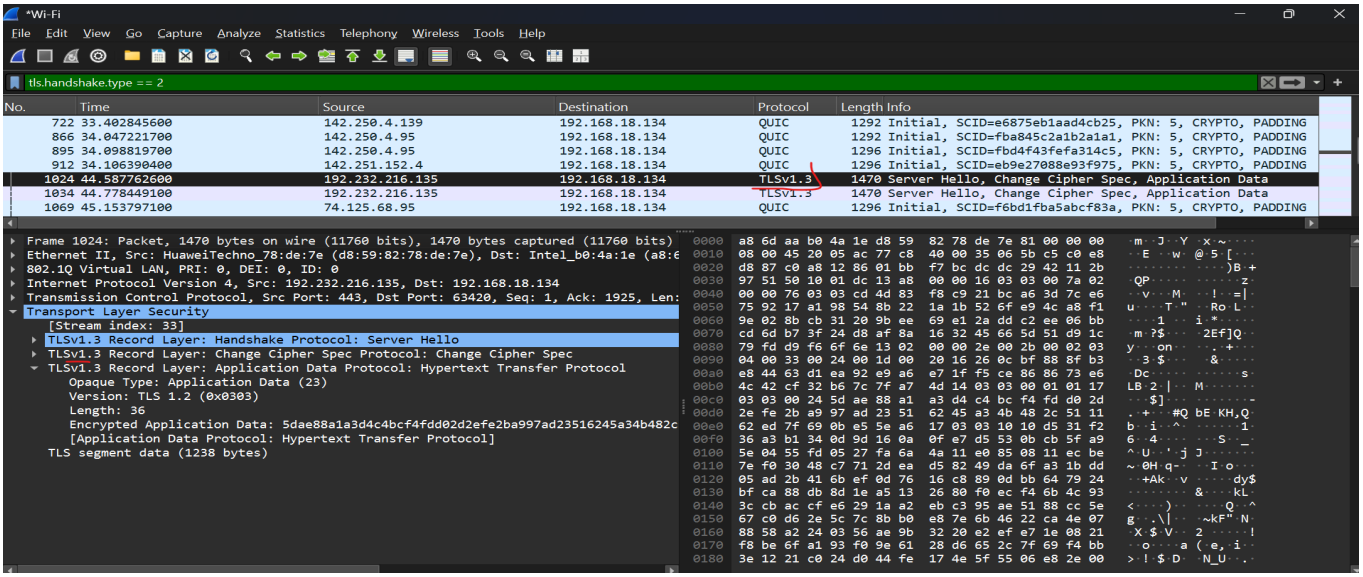
STEP 6

Apply this filter:

```
tls.handshake.type == 2
```

This shows the Server Hello, the reply from the website. Pick the one where the Source is the website IP from Step 4.

Look at the Protocol column. It shows the TLS version.



Answer6: TLS version is TLSv1.3

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